

CALIFORNIA STATE UNIVERSITY, SAN BERNARDINO
Jack H. Brown College of Business and Public Administration
School of Cyber and Decision Sciences

CLASS INFORMATION:

Semester: Fall 2026
Subject: SCM/IST 3500
Title: Problem Solving and Decision-Making
Units: 3
Instruction Mode: Fully Online

INSTRUCTOR'S INFORMATION:

Name: Dr. Lewis A. Njualement
Lewis.Njualement@csusb.edu
Class Meeting Time in Vietnam Tuesday 7.00 am - 9.25 am
Class Meeting Time: Monday 5.00 pm PST – 7.25 pm PST

TEXTBOOK:

The text for the course is Business Analytics, 5th Edition by Jeffrey D. Camm, James J. Cochran, Michael J. Fry, Jeffrey W. Ohlmann. 2024. ISBN: 9780357902202

SOFTWARE:

Microsoft Excel

COURSE CATALOG DESCRIPTION:

This course covers many approaches in solving business problems from a managerial point of view. Students experiment with typical techniques and learn why/how these techniques work. Spreadsheets are used to experiment with such techniques. Emphasis is placed on input requirements and interpretation of results. Topics include descriptive analytics, linear regression, forecasting techniques, spreadsheet modelling, and decision trees.

COURSE DETAILS:

Decision making is an essential skill for managers as they oversee the entire range of activities for efficient and effective delivery of an organization's products or services to its customers. Many long- and short-term decisions are made every day up and down the global businesses. Capturing and analyzing relevant data help organizations to make fact-based decisions to improve their performance and maximize their profit. This course introduces how to use different tools and methods to solve a business problem with an emphasis on supply chain management. It also discusses common decision-making scenarios generations such as sensitivity analysis, decision trees, and what-if analysis to help effective decision making for an uncertain condition. Students will learn how to summarize important aspects of large available business datasets, build analytical models, and report outcomes to help organizations increase profit, reduce costs and manage their operations more efficiently.

PROGRAM LEARNING OBJECTIVES:

- **Specific Knowledge and Skills:** Obtain specialized knowledge of value creation and improvement through business operations and supply chain management.
- **Communication Skills:** Communicate effectively orally, written, and digitally, while demonstrating strong interpersonal and relationship skills that motivate, inspire, and empower others, as well as create a collaborative environment.
- **Managerial and Ethical Reasoning:** Understand and value cultural diversity and global environments in finding solutions for logistics and supply chain problems. Be able to demonstrate integrity, empathy, confidence, and leadership in employing socially responsible and sustainable actions in strategic and operational logistics and supply chain planning that supports ethical business practices.
- **Problem Solving:** Be able to use information technology, apply analytical and managerial concepts and methodologies, and other tools for data analysis, in problem solving, decision-making, planning, and execution of processes and activities in organizations and supply chains.

COURSE LEARNING OBJECTIVES:

CLO1: Explain analytics as a powerful tool for solving business problems and identify the role of data in making fact-based decisions.

CLO2: Use Excel to implement descriptive analytics methods and transfer business data into valuable insights.

CLO3: Outline the outcome of applied analytics by using data visualization techniques to communicate the ideas with the stakeholders and managers for effective decision making.

CLO4: Work as part of a team to generate decisions scenarios to solve business problems and showcase the information to a broad audience through storytelling.

CLASS SESSIONS:

This is a fully online course. Therefore, there will be synchronous online sessions via TEAMS during scheduled class meetings.

COURSE STRUCTURE:

The class uses a modular learning design, meaning that you will see a module for each week in the LMS. You need to follow the tasks you have been assigned for each week and finish them as scheduled. You will develop new skills step by step during weekly modules. Therefore, you need to complete the weekly tasks on time to get ready for the upcoming week.

ASSIGNMENTS:

- **Homework Assignments:**

There will be an individual homework assignment every week on this course. Students should study the assigned material, then complete the homework. Problem solving and decision making with analytics is a subject that cannot be learned merely by listening to lectures -- you must jump in and work on examples to understand the material. If you have trouble with a particular example, ***I am here to help! Don't hesitate to reach out to me by email.*** Also, be sure to pay attention to the due dates of homework assignments!

CASES:

There will be 3 cases for the course. The cases are more in-depth (and real-life scenarios) than the examples you are working on for homework. They are challenging, but interesting and important! And they give you a chance to practice your communication skills.

GRADING:

(A) Grade Distribution:

Your course grade will be determined as follows:

	<u>Percentage</u>
Homework Assignments	40% of total grade
Cases	60% of total grade
<u>Total</u>	100%

(B) Policies:

- **Deadlines:** Late submissions will receive %50 points deduction. If you have a verifiable medical or other special condition that interfere with your work, please let me know ASAP.

(C) Scale:

I will use the following + and - scale:

- $90 \leq A- < 95$ | $A \geq 95$
- $80 \leq B- < 83$ | $83 \leq B < 87$ | $87 \leq B+ < 90$
- $70 \leq C- < 73$ | $73 \leq C < 77$ | $77 \leq C+ < 80$
- $60 \leq D- < 63$ | $63 \leq D < 67$ | $67 \leq D+ < 70$
- $F < 60$

Written Communication: All written assignments should be professional; completed with clarity and purpose, using well-reasoned arguments (with relevant, supportive examples), clearly organized and well-formatted (easy to understand and follow). Written communications and/or assignments that fail to meet these standards will lose points.

Accommodation: In compliance with university policy and equal access laws, the instructor is available to discuss appropriate academic accommodations that may be required for students with disabilities. Requests for academic accommodation are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. If you need accommodation, please see me and contact Services to Students with Disabilities (SSD) office (909-537-5238). Here is the link to their services (<http://ssd.csusb.edu/>).

Policy on plagiarism (From CSUSB's 2021 – 2022 Catalog)

Plagiarism and cheating are violations of the *Standards for Student Conduct* (Title 5, §41301, California Code of Regulations). Definitions and procedures for addressing cheating and plagiarism are found in the Faculty Senate's Policies and Procedures Concerning Academic Dishonesty (FAM 803.5) and Executive Order 1098-Revised 3/29/2019 - Student Conduct Procedures, and may be addressed by both the instructor and the Student Conduct Administrator.

Procedures for addressing cheating and plagiarism are described below. Questions about academic dishonesty and the policy should be addressed to the Office of Student Conduct and Ethical Development (student-conduct@csusb.edu).

Plagiarism is the act of presenting the ideas and writing of another as one's own. Cheating is the act of obtaining or attempting to obtain credit for academic work through the use of any dishonest, deceptive, or fraudulent means.

Cheating includes but is not limited to:

- ☐ Copying, part or in whole, from another's test, software, or other evaluation instrument;
- ☐ Submitting work previously graded in another course unless this has been approved by the

- ☐ course instructor or by departmental policy;
- ☐ Submitting work simultaneously presented in two courses, unless this has been approved by both course instructors or by the department policies of both departments;
- ☐ Using or consulting during examination sources or materials not authorized by the instructor;
- ☐ Altering or interfering with grading or grading instructions;
- ☐ Sitting for an examination by a surrogate, or as a surrogate;

- ☐ Any other act committed by a student during his or her academic work, which defrauds or misrepresents, including aiding or abetting in any of the actions defined above.

Plagiarism is academically dishonest and subjects the offending student to penalties up to and including expulsion. Students must make appropriate acknowledgements of the original source where material written or compiled by another is used.

The instructor may employ any of the following sanctions:

- ☐ Verbal or written reprimand;
- ☐ Assignment of appropriate tasks or examination;
- ☐ Change of grade, including assigning a punitive grade to the work involving dishonesty, or for the course, project, thesis, or any other summary evaluation of the student's academic work.

If the student does not wish to accept the sanction proposed by the instructor, the student may request and require that the allegation be referred to the Office of Student Conduct and Ethical Development. In that event, the procedures specified under Executive Order 1098-Revised 3/29/2019 – Student Conduct Procedures of the California State University shall be observed. The instructor shall not impose any sanctions other than the sanction(s) imposed through the disciplinary procedure.

Student Conduct: Students are afforded reasonable rights as productive individuals of the campus community to function in an environment conducive to learning and personal growth. It is the responsibility of every student to ensure that their actions do not interfere with someone else's right to be a productive individual on campus. Students who violate the Student Conduct Code will be referred to the Office of Student Conduct and Ethical Development at (909) 537-7172. Here is the website link for more information (<http://studentaffairs.csusb.edu/studentconduct/index.html>).

Equity, Diversity, and Inclusion: I am committed to the values of inclusion, diversity, and equity. Here at CSUSB we are committed to creating an inclusive learning environment. If you experience any barriers to learning in this course, please discuss your concerns with me.

Syllabus Changes: The instructor may elect to change the content of the syllabus to better accommodate the learning needs of the class. Students will be notified of any changes.

COURSE OUTLINE – Tentative Schedule:

The following course outline is meant as a general guide only. It may be adjusted based on the students' understanding of the material. Any changes will be announced in LMS.

Module	LECTURE TOPICS	Study/ Textbook Reading	Class Assignments
Week 1: 08/03 Chapter 1	– Introduction to the course – Section 1.1 through 1.3	Textbook: Chapter 1	
Week 2: 08/10 Chapter 1	– Introduction to the course – Section 1.4 through 1.6	Textbook: Chapter 1	Homework Assignment #1 due August 16th at 11.59PM
Week 3: 08/17 Chapter 2	– Descriptive statistics – Section 2.1 through 2.4	Textbook: Chapter 2 Section 2.1 through 2.4	
Week 4: 08/24 Chapter 2	– Descriptive statistics – Section 2.5 through 2.9	Textbook: Chapter 2 Section 2.5 through 2.9	Homework Assignment #2 due August 30th at 11.59PM Case # 1 assigned.
Week 5: 08/31 Chapter 8	– Linear regression – Section 8.1 through 8.3	Textbook: Chapter 8 Section 8.1 through 8.3	Case # 1 due September 6 th at 11.59PM
Week 6: 09/07 Chapter 8	– Linear regression – Section 8.4 through 8.5	Textbook: Chapter 8 Section 8.4 through 8.5	Homework Assignment #3 due September 13 th at 11.59PM
Week 7: 09/14 Chapter 9	– Time series analysis and forecasting – Section 9.1 through 9.3	Textbook: Chapter 9 Section 9.1 through 9.3	
Week 8: 09/21 Chapter 9	– Time series analysis and forecasting – Section 9.4 through 9.5	Textbook: Chapter 9 Section 9.4 through 9.5	Homework Assignment #4 due September 27 th at 11.59PM Case # 2 assigned.

Week 9: 09/28 Chapter 12	– Spreadsheet models – Section 12.1 through 12.3	Textbook: Chapter 12 Section 12.1 through 12.3	Case # 2 due October 4 th at 11.59PM
Week 10: 10/05 Chapter 12	– Spreadsheet models – Section 12.4 through 12.5	Textbook: Chapter 12 Section 12.4 through 12.5	Homework Assignment #5 due October 11 th at 11.59PM
Week 11: 10/12 Chapter 17	– Decision analysis – Section 17.1 through 17.2	Textbook: Chapter 17 Section 17.1 through 17.2	
Week 12: 10/19 Chapter 17	– Decision analysis – Section 17.3 through 17.4	Textbook: Chapter 17 Section 17.3 through 17.4	Homework Assignment #6 due October 25 th at 11.,59PM
Week 13: 10/26 Review	– General review – Data & ERP – Analytics & AI	Discussion Week	Case # 3 (Final Exam) assigned
Final Exam	– Final Exam		Case # 3 (Final Exam) due on November 22nd